"e"

Exam code: 9MA0

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Written by: Paul
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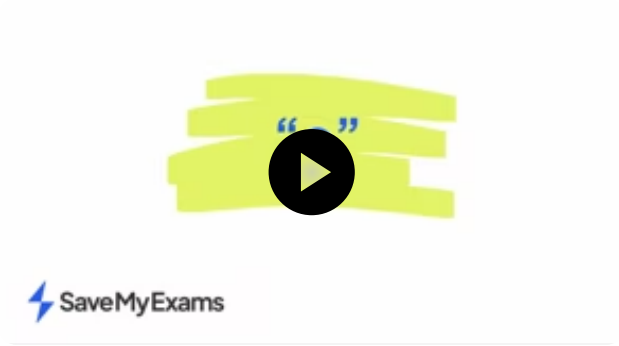

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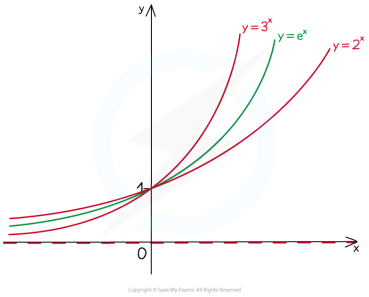
"e"

What is the number "e" in maths?

- **e** is an irrational number, sometimes called Euler's number
  - **e ≈ 2.718**

How do I sketch y = e^x?

- The exponential function is **y = e^x**
  - As with other exponential graphs **y = e^x**
    - passes through (0, 1)
    - has the x-axis as an asymptote



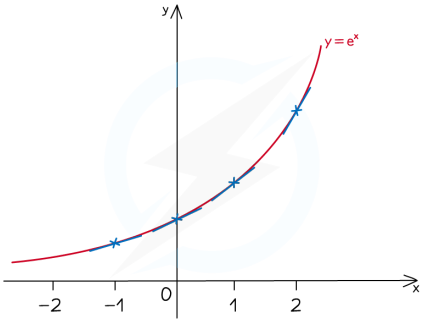
What is the special property of "e"?

- **y = e^x** has the particular property that

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- ie for *every* real number **x**, the gradient of $\mathbf{y = e^x}$ is also equal to $\mathbf{e^x}$ (see Derivatives of Exponential Functions)



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x	y	dy / dx
−1	0.3678...	0.3678...
0	1	1
1	2.7182...	2.7182...
2	7.3890...	7.3890...

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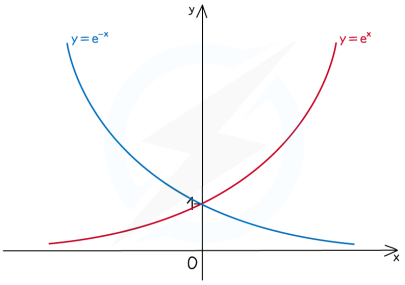
How do I sketch $y = e^{-x}$?

- $\mathbf{y = e^{-x}}$ is a reflection in the **y**-axis of $\mathbf{y = e^x}$
- They are of the form **y = f(x)** and **y = f(−x)** (see Transformations of Functions - Reflections)



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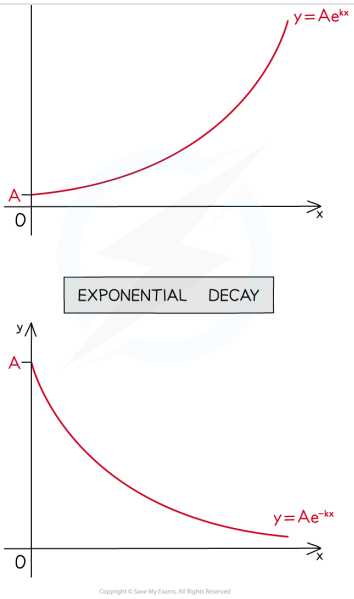
What is exponential growth and decay?



- **y = Ae^{kx}** (**k > 0**) is exponential growth
- **y = Ae^{-kx}** (**k > 0**) is exponential decay
- **A** is the initial value
- **k** is a (usually positive) constant
- "–" is used in the equation making clear whether it is growth or decay

Worked Example

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